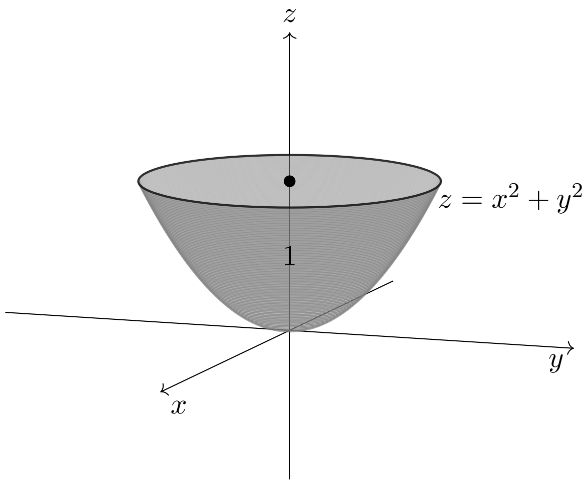
1. Calculate the volume this shape encloses, in all the three coordinate systems. The height of this shape is 1 unit along the z-axis. Comment on which system gives simpler calculations and why that is the case.



Solution:
The region is bounded above by $z = 1$ and below by the paraboloid $z = x^2 + y^2$.

Q1) Cartiesian:

We have to integrate the volume element in the cartesian system $dxdydz$ over the appropriate limits.

$$\iiint_V dxdydz$$

If we keep y and z constant, x varies from $-\sqrt{z-y^2}$ to $\sqrt{z-y^2}$. Then, for a constant z, y varies from $-\sqrt{z}$ to $\sqrt{z}$ and finally z varies from 0 to 1.

$$\boxed{V = \int_0^1 \int_{-\sqrt{z}}^{\sqrt{z}} \int_{-\sqrt{z-y^2}}^{\sqrt{z-y^2}} dxdydz}$$

solving this integral

$$\Rightarrow \int_0^1 \int_{-\sqrt{z}}^{\sqrt{z}} \int_{-\sqrt{z-y^2}}^{\sqrt{z-y^2}} dxdydz$$

$$\Rightarrow 2\int_0^1 \int_{-\sqrt{z}}^{\sqrt{z}} \sqrt{z-y^2}\; dydz$$

Here let $y = \sqrt{z}\sin\theta$

$$\Rightarrow 2\int_0^1 \int_{-\sqrt{z}}^{\sqrt{z}} \sqrt{z - y^2}\ dydz \quad \longrightarrow \quad 2\int_0^1 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} z\cos^2\theta\ d\theta dz$$

$$\Rightarrow \frac{\pi}{2}\int_0^1 zdz$$

$$\Rightarrow \boxed{V = \frac{\pi}{2}}$$

Cylindrical:

We have to integrate the volume element in the cylindrical system $dr\ rd\theta\ dz$

$$\iiint rdrd\theta dz$$

If we keep θ and z constant, r varies from 0 to $\sqrt{z}$, Then if keep z, θ varies from 0 to 2π and finally z varies from 0 to 1.

$$\boxed{V = \int_0^1 \int_0^{2\pi} \int_0^{\sqrt{z}} rdrd\theta dz}$$

solving this integral

$$\Rightarrow \int_0^1 \int_0^{2\pi} \int_0^{\sqrt{z}} rdrd\theta dz$$

$$\Rightarrow \int_0^1 \int_0^{2\pi} \frac{z}{2} d\theta dz$$

$$\Rightarrow \pi \int_0^1 zdz$$

$$\Rightarrow \boxed{V = \frac{\pi}{2}}$$

Spherical:

We have to integrate the volume element in the spherical system $d\rho\ \rho d\phi\ \rho\sin\phi d\theta$

$$\iiint \rho^2 \sin\phi\ d\rho d\phi d\theta$$

you can try to figure out the limits for this but you need to solve transcendental equations.

Based on the solutions, it is obvious that the cylindrical system is more suited for this question, because this shape is axially symmetric .

2. In spherical coordinates (r, θ, ϕ) (θ is the polar angle), a particle follows the path given by

$$r(t) = 1,\ \theta(t) = \frac{\pi}{4} + 0.5t,\ \phi(t) = 4t$$

Compute its velocity and acceleration vectors at $t = 0$

solution:
In spherical coordinates, the position of any point can be represented by the vector $\vec{r} = r\hat{e_r}$. The velocity and acceleration can be found by taking the first and second time derivatives of this vector.

The key point to note here is that in curvilinear coordinates, the basis vectors are not constant, so we need to also take the derivatives of the basis vectors.

We can represent the spherical basis in cartesian coordinates as

$$\begin{aligned}
\hat{e_r} &= \sin\theta\cos\phi\hat{i} + \sin\theta\sin\phi\hat{j} + \cos\theta\hat{k} \\
\hat{e_\theta} &= \cos\theta\cos\phi\hat{i} + \cos\theta\sin\phi\hat{j} - \sin\theta\hat{k} \\
\hat{e_\phi} &= -\sin\phi\hat{i} + \cos\phi\hat{j}
\end{aligned}$$

taking the derivatives of $\hat{e_r}$, we get

$$\begin{aligned}
\frac{d\hat{e_r}}{dr} &= \frac{d(\sin\theta\cos\phi\hat{i} + \sin\theta\sin\phi\hat{j} + \cos\theta\hat{k})}{dr} = 0 \\
\frac{d\hat{e_r}}{d\theta} &= \frac{d(\sin\theta\cos\phi\hat{i} + \sin\theta\sin\phi\hat{j} + \cos\theta\hat{k})}{d\theta} = \cos\theta\cos\phi\hat{i} + \cos\theta\sin\phi\hat{j} - \sin\theta\hat{k} = \hat{e_\theta} \\
\frac{d\hat{e_r}}{d\phi} &= \frac{d(\sin\theta\cos\phi\hat{i} + \sin\theta\sin\phi\hat{j} + \cos\theta\hat{k})}{d\phi} = -\sin\theta\sin\phi\hat{i} + \sin\theta\cos\phi\hat{j} = \sin\theta\hat{e_\phi}
\end{aligned}$$

So we get

$$d\hat{e_r} = d\theta\hat{e_\theta} + d\phi\sin\theta\hat{e_\phi}$$

Similarly, the derivatives of the other two basis vectors can also be calculated, which gives

$$d\hat{e_r} = d\theta\vec{e_\theta} + d\phi\sin\theta\vec{e_\phi}$$

$$d\hat{e_\theta} = -d\theta\vec{e_r} + d\phi\cos\theta\vec{e_\phi}$$
$$d\hat{e_\phi} = -d\phi\sin\theta\vec{e_r} - d\phi\cos\theta\vec{e_\theta}$$

Using this, we get

$$\vec{v} = \frac{d\vec{r}}{dt} = r\frac{d\hat{e_r}}{dt}$$
$$\Rightarrow \vec{v} = \frac{d\theta}{dt}\hat{e_\theta} + \frac{d\phi}{dt}\sin\theta\vec{e_\phi}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$
$$\Rightarrow \frac{d^2\theta}{dt^2}\hat{e_\theta} + \frac{d\theta}{dt}\frac{d\hat{e_\theta}}{dt} + \frac{d^2\phi}{dt^2}\sin\theta\vec{e_\phi} + \frac{d\phi}{dt}\frac{d\theta}{dt}\cos\theta\vec{e_\phi} + \frac{d\phi}{dt}\sin\theta\frac{d\hat{e_\phi}}{dt}$$
$$\Rightarrow \left(-\left(\frac{d\theta}{dt}\right)^2 - \left(\frac{d\phi}{dt}\right)^2\sin^2\theta\right)\hat{e_r} + \left(\frac{d^2\theta}{dt^2} - \left(\frac{d\phi}{dt}\right)^2\sin\theta\cos\theta\right)\hat{e_\theta} + \left(2\frac{d\theta}{dt}\frac{d\phi}{dt}\cos\theta + \frac{d^2\phi}{dt^2}\sin\theta\right)\hat{e_\phi}$$

substituting the values of the derivatives $\frac{d\theta}{dt} = 0.5$, $\frac{d\phi}{dt} = 4$, we get

$$\boxed{\begin{aligned}\vec{v} &= 0.5\hat{e_r} + 2\sqrt{2}\hat{e_\phi}\\ \vec{a} &= -\frac{33}{4}\hat{e_r} - 8\hat{e_\theta} + 2\sqrt{2}\hat{e_\phi}\end{aligned}}$$

3. Consider the vector field in $\mathbb{R}^3$

$$\mathbf{V}(\mathbf{r}) = \frac{\hat{\mathbf{r}}}{r^2} = \frac{\mathbf{r}}{r^3}, \qquad r = \|\mathbf{r}\|,$$

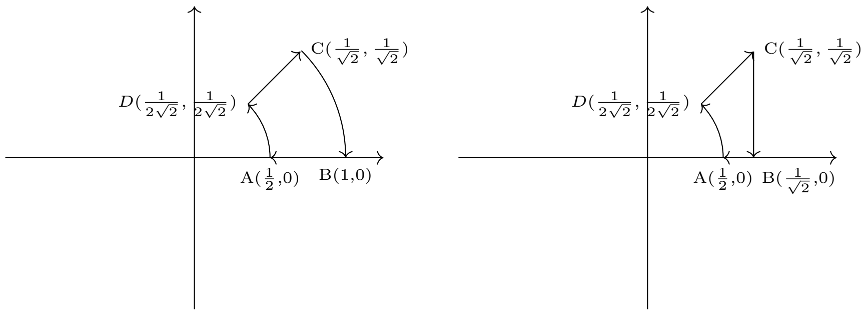
defined for all $\mathbf{r} \neq \mathbf{0}$. The field is singular at the origin and is often interpreted as the velocity field of an idealized point source (radial outflow) in an incompressible fluid.

(a) **Show that** $\nabla \cdot \mathbf{V}(\mathbf{r}) = 0$ for every $\mathbf{r} \neq \mathbf{0}$.
Explain physically why the divergence vanishes away from the origin, even though the flow appears to "spread out" radially.
Reconcile this with the fact that there is a net outward flux through any sphere centered at the origin (the sphere of radius R):

$$\oint_{S_R} \mathbf{V} \cdot d\mathbf{A} \neq 0,$$

and state what this implies about the divergence at $\mathbf{r} = \mathbf{0}$ (in the distributional sense). [Hint: Use divergence theorem on the sphere of radius R to understand the paradox.]

(b) Consider the same vector field as above and compute the closed loop line integral over the following 2 paths and note the observations. Also use the stokes theorem to compute the same and try to understand what the specialty of this particular vector field is.



Solution:

(a): Radial Fluid Outflow

Given: Velocity field $\mathbf{V}(\mathbf{r}) = \dfrac{\hat{r}}{r^2}$ for $r \neq 0$.

Why is $\nabla \cdot \mathbf{V} = 0$ away from the origin?

In spherical coordinates, for a radial field $\mathbf{V} = V_r(r)\hat{r}$:

$$\nabla \cdot \mathbf{V} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 V_r)$$

For $V_r = \frac{1}{r^2}$:

$$\nabla \cdot \mathbf{V} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \cdot \frac{1}{r^2}\right) = \frac{1}{r^2}\frac{\partial}{\partial r}(1) = 0$$

This holds for all $r > 0$. The singularity is located at $\mathbf{r} = \mathbf{0}$ (the origin).

Net Volume Outflow from Sphere of Radius R

We compute the flux integral directly through the spherical surface:

$$\Phi = \iint_S \mathbf{V} \cdot d\mathbf{S}$$

On a sphere of radius R, the outward normal is $\hat{n} = \hat{r}$ and $d\mathbf{S} = R^2 \sin\theta \, d\theta \, d\phi \, \hat{r}$.

The velocity field at $r = R$ is:

$$\mathbf{V}(R) = \frac{\hat{r}}{R^2}$$

Therefore:

$$\mathbf{V} \cdot d\mathbf{S} = \frac{\hat{r}}{R^2} \cdot \hat{r}\, R^2 \sin\theta\, d\theta\, d\phi = \sin\theta\, d\theta\, d\phi$$

Integrating over the sphere ($\theta \in [0, \pi]$, $\phi \in [0, 2\pi]$):

$$\Phi = \int_0^{2\pi} \int_0^{\pi} \sin\theta\, d\theta\, d\phi = \int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta\, d\theta$$

$$= 2\pi \cdot [-\cos\theta]_0^{\pi} = 2\pi \cdot [-(-1) - (-1)] = 2\pi \cdot 2 = 4\pi$$

Answer: The net volume outflow is $\boxed{4\pi}$.

Why doesn't the flux vanish?

Using divergence theorem directly without accounting for the point of singularity we get:

$$\iiint_{volume} (\nabla.\mathbf{V}) dx \cdot dy \cdot dz = \oiint_{surface} \mathbf{V} \cdot d\mathbf{S}$$

Apparently it seems that

$$LHS = 0$$

$$RHS = 4\pi$$

Although $\nabla \cdot \mathbf{V} = 0$ at every point away from the origin, the velocity field itself has a singularity at $r = 0$. The local expansion rate being zero tells us about the behavior at regular points in the field, but cannot account for what happens at the singular point.

At the origin, the field is undefined and diverges to infinity. This singularity acts as a source of fluid. When we integrate the flux through any closed surface enclosing the origin, we capture the total output from this point source, which is finite (4π) despite the field being well-behaved everywhere else.

The key insight is that pointwise properties (like $\nabla \cdot \mathbf{V} = 0$ at regular points) don't necessarily determine global properties (like total flux through a closed surface) when singularities are present inside the surface. Here if we want to know the functional form of the divergence it is a 3 dimensional dirac delta

$$\nabla.V = 4\pi\delta^3(x, y, z)$$

such that when you integrate this function over any volume enclosing origin we get:

$$\iiint_{volume} 4\pi\delta^3(x, y, z) dx \cdot dy \cdot dz = 4\pi$$

For more information on Dirac delta (not a practical function) refer:click here

(b) Line Integrals of Radial Field

For the 2D projection, $\mathbf{V} = \frac{\hat{r}}{r^2} = \frac{x\hat{\imath} + y\hat{\jmath}}{(x^2+y^2)^{3/2}}$.

Path 1: $A \to B \to C \to D \to A$

Points: $A(\frac{1}{2}, 0)$, $B(1, 0)$, $C(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, $D(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}})$.

Segment $A \to B$**:** Along $y = 0$, $x : \frac{1}{2} \to 1$

$$\int_A^B \mathbf{V} \cdot d\mathbf{l} = \int_{1/2}^1 \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_{1/2}^1 = -1 + 2 = 1$$

Segment $B \to C$**:** Radial path from $r = 1$ to $r = 1$ at angle $\theta : 0 \to \frac{\pi}{4}$. In polar coordinates, $\mathbf{V} = \frac{\hat{r}}{r^2}$ and $d\mathbf{l} = dr\,\hat{r} + r\,d\theta\,\hat{\theta}$.

$$\mathbf{V} \cdot d\mathbf{l} = \frac{1}{r^2} dr$$

Since r is constant ($dr = 0$):

$$\int_B^C \mathbf{V} \cdot d\mathbf{l} = 0$$

Segment $C \to D$**:** Along $y = x$, from $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ to $(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}})$. Parameterize: $x = t$, $y = t$, $t : \frac{1}{\sqrt{2}} \to \frac{1}{2\sqrt{2}}$.

$$\mathbf{V} = \frac{t\hat{\imath} + t\hat{\jmath}}{(2t^2)^{3/2}} = \frac{t(\hat{\imath} + \hat{\jmath})}{2\sqrt{2}t^3} = \frac{\hat{\imath} + \hat{\jmath}}{2\sqrt{2}t^2}$$

$$d\mathbf{l} = dt(\hat{\imath} + \hat{\jmath})$$

$$\mathbf{V} \cdot d\mathbf{l} = \frac{2}{2\sqrt{2}t^2} dt = \frac{1}{\sqrt{2}t^2} dt$$

$$\int_C^D \mathbf{V} \cdot d\mathbf{l} = \frac{1}{\sqrt{2}} \int_{1/\sqrt{2}}^{1/(2\sqrt{2})} \frac{1}{t^2} dt = \frac{1}{\sqrt{2}} \left[-\frac{1}{t}\right]_{1/\sqrt{2}}^{1/(2\sqrt{2})}$$

$$= \frac{1}{\sqrt{2}} \left[-2\sqrt{2} + \sqrt{2}\right] = \frac{1}{\sqrt{2}}(-\sqrt{2}) = -1$$

Segment $D \to A$**:** Circular arc at $r = \frac{1}{2}$, $\theta : \frac{\pi}{4} \to 0$ (similar to $B \to C$):

$$\int_D^A \mathbf{V} \cdot d\mathbf{l} = 0$$

Total for Path 1:

$$\oint \mathbf{V} \cdot d\mathbf{l} = 1 + 0 - 1 + 0 = \boxed{0}$$

Path 2: $A \to B' \to C \to D \to A$

Points: $A(\frac{1}{2}, 0)$, $B'(\frac{1}{\sqrt{2}}, 0)$, $C(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, $D(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}})$.

Segment $A \to B'$**:** Along $y = 0$, $x : \frac{1}{2} \to \frac{1}{\sqrt{2}}$

$$\int_A^{B'} \mathbf{V} \cdot d\mathbf{l} = \left[-\frac{1}{x}\right]_{1/2}^{1/\sqrt{2}} = -\sqrt{2} + 2$$

Segment $B' \to C$**:** Along $x = \frac{1}{\sqrt{2}}$, $y : 0 \to \frac{1}{\sqrt{2}}$. Here $d\mathbf{l} = dy\,\hat{j}$.

$$\mathbf{V} \cdot d\mathbf{l} = \frac{y}{(1/2 + y^2)^{3/2}} dy$$

$$\int_{B'}^{C} = \int_0^{1/\sqrt{2}} \frac{y}{(1/2 + y^2)^{3/2}} dy$$

Let $u = \frac{1}{2} + y^2$, $du = 2y\,dy$:

$$= \frac{1}{2} \int_{1/2}^{1} u^{-3/2} du = \frac{1}{2} \left[-2u^{-1/2}\right]_{1/2}^{1} = -\left[1 - \sqrt{2}\right] = \sqrt{2} - 1$$

Segments $C \to D$ **and** $D \to A$**:** Same as Path 1: -1 and 0.

Total for Path 2:

$$\oint \mathbf{V} \cdot d\mathbf{l} = (2 - \sqrt{2}) + (\sqrt{2} - 1) - 1 + 0 = \boxed{0}$$

Observation

Both closed loop integrals are zero. This is because $\mathbf{V} = \frac{\hat{r}}{r^2}$ can be written as the gradient of a scalar function $\phi = -\frac{1}{r}$ in the region away from the origin. For such a gradient field, line integrals between two points depend only on the endpoints, not the path. Since both paths are closed loops (same start and end point), the circulation vanishes. This comes from the direct application of stokes theorem which states

$$\iint_{surface} (\nabla \times \mathbf{V}) \cdot d\vec{\mathbf{S}} = \oint_{boundary\ loop} \mathbf{V} \cdot d\vec{\mathbf{l}}$$

so if the vector field is a gradient field of some scalar function then

$$\iint_{surface} (\nabla \times \nabla\Phi) \cdot d\vec{\mathbf{S}} = \oint_{boundary\ loop} \mathbf{V} \cdot d\vec{\mathbf{l}}$$

rearranging

$$\oint_{boundary\ loop} \mathbf{V} \cdot d\vec{\mathbf{l}} = 0$$

4. The temperature distribution in 3D space is given by $T(x, y, z) = x + y + z$
Constraint Surface: A thin metal sheet occupies the ellipsoid surface $g(x, y, z)=\frac{x^2}{4} + y^2 + z^2 - 1 = 0$
Find the points on the metallic surface where it is the hottest and coldest respectively.

Solution:CONSTRAINED OPTIMIZATION

Objective: Maximize/minimize $T(x, y, z) = x + y + z$ subject to $g(x, y, z) = \frac{x^2}{4} + y^2 + z^2 - 1 = 0$.
Use Lagrange multipliers: $\nabla T = \lambda \nabla g$.

$$\nabla T = (\hat{\imath}, \hat{\jmath}, \hat{k}), \quad \nabla g = \left(\frac{x}{2}, 2y, 2z\right)$$

Setting $\nabla T = \lambda \nabla g$:

$$1 = \lambda \frac{x}{2}, \quad 1 = 2\lambda y, \quad 1 = 2\lambda z$$

From these:

$$x = \frac{2}{\lambda}, \quad y = \frac{1}{2\lambda}, \quad z = \frac{1}{2\lambda}$$

Substitute into constraint:

$$\frac{1}{\lambda^2} + \frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} = 1$$

$$\frac{1}{\lambda^2}\left(1 + \frac{1}{4} + \frac{1}{4}\right) = 1 \implies \frac{3}{2\lambda^2} = 1 \implies \lambda^2 = \frac{3}{2} \implies \lambda = \pm\sqrt{\frac{3}{2}}$$

For $\lambda = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2}$:

$$x = \frac{2\sqrt{6}}{3}, \quad y = z = \frac{\sqrt{6}}{6}$$

$$T = \frac{2\sqrt{6}}{3} + \frac{\sqrt{6}}{6} + \frac{\sqrt{6}}{6} = \frac{2\sqrt{6}}{3} + \frac{\sqrt{6}}{3} = \sqrt{6}$$

For $\lambda = -\frac{\sqrt{6}}{2}$:

$$x = -\frac{2\sqrt{6}}{3}, \quad y = z = -\frac{\sqrt{6}}{6}$$

$$T = -\sqrt{6}$$

Answer:

- Hottest point: $\left(\frac{2\sqrt{6}}{3}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}\right)$ with $T = \boxed{\sqrt{6}}$
- Coldest point: $\left(-\frac{2\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right)$ with $T = \boxed{-\sqrt{6}}$

5. A room has a non-uniform temperature distribution given in cylindrical coordinates by

$$T(\rho, \phi, z) = 300 + 2z \quad \text{K}.$$

Inside the room, a spiral staircase of constant radius $r = 2$ m connects the ground floor at $z = 0$ to the top floor at $z = 10$ m. The staircase makes 5 complete turns while ascending. A person carries a block of ice along the staircase from the ground floor to the top floor. At the starting point, the ice is at its melting temperature 0°C. The ice melts according to the model

$$\frac{dm}{ds} = k\big[T(\rho, \phi, z) - 273\big],$$

where $k = 5 \times 10^{-4}$ kg/(m·K), m is the mass of ice, and s is the physical path length along the staircase. Assume that ice remains at 273 K throughout the motion, heat transfer depends only on the local temperature and walking speed is irrelevant. If the mass of ice remaining at the top floor must be exactly 1 kg, determine the initial mass of ice that must be carried from the ground floor.

Solution:
The staircase makes 5 complete turns, so the angular coordinate varies as

$$\phi \in [0, 10\pi].$$

The total vertical rise is 10 m, hence

$$z = \frac{10}{10\pi}\phi = \frac{\phi}{\pi}.$$

In cylindrical coordinates,

$$ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2.$$

As the radius is constant ($\rho = 2$ m), $d\rho = 0$, and therefore

$$ds^2 = 4\, d\phi^2 + dz^2.$$

Using $dz = \frac{1}{\pi} d\phi$,

$$ds^2 = \left(4 + \frac{1}{\pi^2}\right) d\phi^2, \qquad ds = \sqrt{4 + \frac{1}{\pi^2}}\, d\phi.$$

The temperature distribution is

$$T(\rho, \phi, z) = 300 + 2z.$$

Along the staircase,

$$T(\phi) = 300 + \frac{2\phi}{\pi}, \qquad T - 273 = 27 + \frac{2\phi}{\pi}.$$

The melting rate is

$$\frac{dm}{ds} = k\,[T(\phi) - 273].$$

$$\implies \frac{dm}{d\phi} = \frac{dm}{ds}\frac{ds}{d\phi} = k\left(27 + \frac{2\phi}{\pi}\right)\sqrt{4 + \frac{1}{\pi^2}}.$$

So total mass melted is

$$\Delta m = k\sqrt{4 + \frac{1}{\pi^2}} \int_0^{10\pi} \left(27 + \frac{2\phi}{\pi}\right) d\phi = k\sqrt{4 + \frac{1}{\pi^2}}\,(370\pi).$$

$$\Delta m \approx 1.18 \text{ kg}.$$

So required initial mass is:

$$m_{\text{initial}} = 1 + \Delta m \approx 2.18 \text{ kg}.$$

6. Consider the region in $\mathbb{R}^3$ defined by

$$2x - y + 2z \le 5.$$

(a) Find the volume of the largest sphere

$$x^2 + y^2 + z^2 = r^2$$

centered at the origin that lies entirely inside this region.

(b) Write the plane as a dot product

$$\vec{n} \cdot \vec{r} = 5, \quad \text{where } \vec{n} = (2, -1, 2),\ \vec{r} = (x, y, z).$$

Use the Cauchy–Schwarz inequality

$$|\vec{n} \cdot \vec{r}| \le |\vec{n}|\,|\vec{r}|$$

to obtain a bound on r.

You will find that Cauchy–Schwarz naturally gives a *lower bound* on $|\vec{r}|$ for points on the plane, but the problem requires an *upper bound* on r for points inside the half–space.

Explain why the inequality must be interpreted in the opposite direction in this problem, and give a geometric interpretation of the equality case.

Solution:

The plane

$$2x - y + 2z = 5$$

has normal vector

$$\vec{n} = (2, -1, 2).$$

The distance of this plane from the origin is

$$d = \frac{|5|}{\|\vec{n}\|} = \frac{5}{\sqrt{2^2 + (-1)^2 + 2^2}} = \frac{5}{3}.$$

For a sphere centered at the origin to lie entirely inside the half–space

$$2x - y + 2z \leq 5,$$

its radius cannot exceed the perpendicular distance from the origin to the plane. Hence,

$$r_{\max} = \frac{5}{3}.$$

Therefore, the maximum volume of the sphere is

$$V_{\max} = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \left(\frac{5}{3}\right)^3 = \frac{500\pi}{81}.$$

(b) The plane can be written in vector form as

$$\vec{n} \cdot \vec{r} = 5, \quad \text{where } \vec{n} = (2, -1, 2),\ \vec{r} = (x, y, z).$$

Applying the Cauchy–Schwarz inequality,

$$|\vec{n} \cdot \vec{r}| \leq \|\vec{n}\|\, \|\vec{r}\|,$$

we obtain

$$5 \leq 3\, \|\vec{r}\| \quad \Rightarrow \quad \|\vec{r}\| \geq \frac{5}{3}.$$

This shows that *every point on the plane* is at a distance at least $\frac{5}{3}$ from the origin. Thus, Cauchy–Schwarz naturally yields a **lower bound on the distance of the plane from the origin**, which is precisely the perpendicular distance of the plane.

However, the sphere must lie entirely *inside* the half–space

$$\vec{n} \cdot \vec{r} \leq 5.$$

Hence, the farthest point of the sphere from the origin in the direction of $\vec{n}$ must not cross the plane. This converts the lower bound on the plane–origin distance into an *upper bound* on the radius of the sphere, namely

$$r \leq \frac{5}{3}.$$

Equality occurs when the radius vector of the sphere is parallel to the normal vector $\vec{n}$. Geometrically, this corresponds to the sphere touching the plane at exactly one point, with the radius at the point of contact being perpendicular to the plane.

7. A vector field is given in spherical coordinates by

$$\vec{F}(r, \theta, \phi) = \frac{\cos\phi}{r^2}\,\hat{r} + \frac{\sin\phi}{r^2 \sin\theta}\,\hat{\phi}.$$

   (a) Plot the magnitude $|\vec{F}|$ as:
      i. a function of ϕ for $r = 1$ and $\theta = \pi/3$,
      ii. a function of r for $\phi = \pi/4$ and $\theta = \pi/2$.
   (b) Compute by hand and visualize them using MATLAB:
      i. the divergence $\nabla \cdot \vec{F}$,
      ii. the curl $\nabla \times \vec{F}$.
   (c) Check whether the vector field $\vec{F}$ is conservative using **two independent methods**, state your final conclusion and clearly specify the region over which your conclusion is valid.

Solution:

The vector field is given in spherical coordinates by

$$\vec{F}(r, \theta, \phi) = \frac{\cos\phi}{r^2}\,\hat{r} + \frac{\sin\phi}{r^2 \sin\theta}\,\hat{\phi}.$$

(a) Magnitude of the Field
The magnitude of a vector field in spherical coordinates is

$$|\vec{F}| = \sqrt{F_r^2 + F_\theta^2 + F_\phi^2}.$$

Here,

$$F_r = \frac{\cos\phi}{r^2}, \qquad F_\theta = 0, \qquad F_\phi = \frac{\sin\phi}{r^2\sin\theta}.$$

Hence,

$$|\vec{F}| = \frac{1}{r^2}\sqrt{\cos^2\phi + \frac{\sin^2\phi}{\sin^2\theta}}.$$

This expression is used directly for plotting as required.

(b)
- **Divergence of the Field**
  The divergence in spherical coordinates is

$$\nabla\cdot\vec{F} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2F_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta F_\theta) + \frac{1}{r\sin\theta}\frac{\partial F_\phi}{\partial\phi}.$$

  Substituting the components:

$$r^2F_r = \cos\phi \quad\Rightarrow\quad \frac{\partial}{\partial r}(r^2F_r) = 0,$$

$$F_\theta = 0 \quad\Rightarrow\quad \frac{\partial}{\partial\theta}(\sin\theta F_\theta) = 0,$$

$$\frac{\partial F_\phi}{\partial\phi} = \frac{\cos\phi}{r^2\sin\theta}.$$

  Therefore,

$$\nabla\cdot\vec{F} = \frac{1}{r\sin\theta}\cdot\frac{\cos\phi}{r^2\sin\theta} = \frac{\cos\phi}{r^3\sin^2\theta}.$$

- **Curl of the Field**
  The curl in spherical coordinates is given by

$$\nabla\times\vec{F} = \begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ F_r & rF_\theta & r\sin\theta F_\phi \end{vmatrix}.$$

  Using

$$rF_\theta = 0, \qquad r\sin\theta F_\phi = \frac{\sin\phi}{r}, \qquad F_r = \frac{\cos\phi}{r^2},$$

  each component of the curl evaluates to zero after differentiation.
  Thus,

$$\boxed{\nabla\times\vec{F} = \vec{0}}$$

  everywhere the field is defined.

(c) **Conservative Field Test — Method 1 (Potential Function)**

We check whether a scalar function $\psi(r, \theta, \phi)$ exists such that

$$\vec{F} = \nabla\psi.$$

The gradient in spherical coordinates is

$$\nabla\psi = \frac{\partial\psi}{\partial r}\,\hat{r} + \frac{1}{r}\frac{\partial\psi}{\partial\theta}\,\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial\psi}{\partial\phi}\,\hat{\phi}.$$

Comparing components:
From the $\hat{r}$ component,

$$\frac{\partial\psi}{\partial r} = \frac{\cos\phi}{r^2} \quad \Rightarrow \quad \psi = -\frac{\cos\phi}{r} + f(\theta, \phi).$$

From the $\hat{\phi}$ component,

$$\frac{1}{r\sin\theta}\frac{\partial\psi}{\partial\phi} = \frac{\sin\phi}{r^2\sin\theta} \quad \Rightarrow \quad \frac{\partial\psi}{\partial\phi} = \frac{\sin\phi}{r}.$$

Differentiating the earlier expression for ψ,

$$\frac{\partial}{\partial\phi}\left(-\frac{\cos\phi}{r}\right) = \frac{\sin\phi}{r},$$

which is consistent. Hence, $f(\theta, \phi)$ is a constant and

$$\boxed{\psi(r, \theta, \phi) = -\frac{\cos\phi}{r} + C}.$$

The $\hat{\theta}$ component of the gradient in spherical coordinates is

$$\frac{1}{r}\frac{\partial\psi}{\partial\theta}\,\hat{\theta}.$$

From the previously obtained scalar potential,

$$\psi(r, \theta, \phi) = -\frac{\cos\phi}{r} + C,$$

we compute the partial derivative with respect to θ:

$$\frac{\partial\psi}{\partial\theta} = \frac{\partial}{\partial\theta}\left(-\frac{\cos\phi}{r} + C\right) \implies \frac{\partial\psi}{\partial\theta} = 0 \implies \boxed{\frac{1}{r}\frac{\partial\psi}{\partial\theta} = 0}$$

which exactly matches the $\hat{\theta}$ component of the given vector field, confirming consistency.

Method 2 (Curl Criterion)
We got

$$\nabla \times \vec{F} = \vec{0}$$

everywhere the field is defined. Therefore, the field is locally conservative. However, the field is undefined at:

$$r = 0 \quad \text{and} \quad \sin\theta = 0 \; (\theta = 0, \pi).$$

$$\boxed{\text{The vector field is conservative in any region excluding } r = 0 \text{ and the axes } \theta = 0, \pi.}$$

The conclusion is valid only within such regions.

8. An Oceanographer measures the surface velocity to be

$$\vec{v}(x, y) = x^2\hat{j}$$

Compute the curl of the velocity field at (2,0). Why is it non zero even though all the velocity vectors point in the same direction?

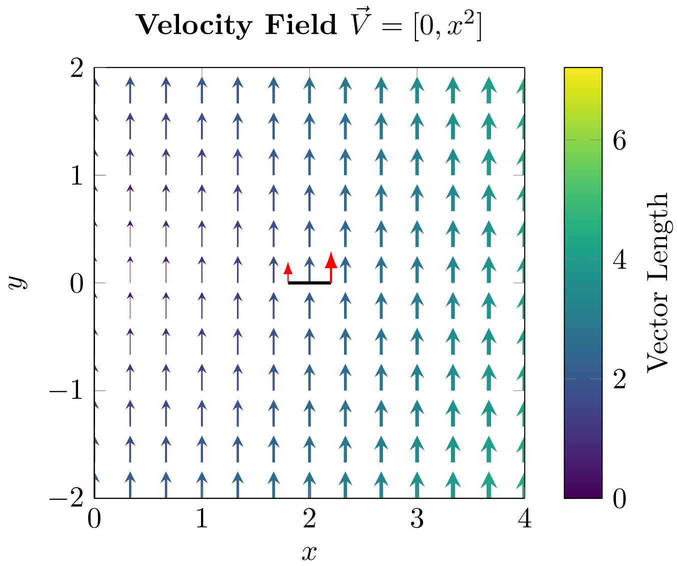
Solution:

The curl of the velocity field is given by

$$\nabla \times \vec{v} = \frac{dv_y}{dx} - \frac{dv_x}{dy}$$

$$\Rightarrow \nabla \times \vec{v} = 2x$$

$$\Rightarrow \nabla \times \vec{v}|_{(2,0)} = 4$$



The curl is non-zero because the magnitude of the velocity vectors vary. Imagine placing a small stick at (2,0), the right side of the stick experiences a greater force than the left side, so there is a net torque on the stick which causes it to rotate counterclockwise.

9. A scalar field is defined in cylindrical coordinates by

$$\Phi(\rho, \phi, z) = \rho^2 z.$$

(a) Compute the gradient $\nabla\Phi$ in cylindrical coordinates.

(b) Evaluate the line integral

$$\int_C \nabla\Phi \cdot d\mathbf{l}$$

where C is the helix defined by

$$\rho = a, \quad \phi = t, \quad z = bt, \quad 0 \le t \le 2\pi.$$

(c) Verify your result using the fundamental theorem of line integrals, and comment on the physical significance.

Solution

A scalar field in cylindrical coordinates is given by

$$\Phi(\rho, \phi, z) = \rho^2 z.$$

(a) Gradient in cylindrical coordinates

Recall the gradient operator in cylindrical coordinates:

$$\nabla\Phi = \hat{\boldsymbol{\rho}}\,\frac{\partial\Phi}{\partial\rho} + \hat{\boldsymbol{\phi}}\,\frac{1}{\rho}\frac{\partial\Phi}{\partial\phi} + \hat{\mathbf{z}}\,\frac{\partial\Phi}{\partial z}.$$

Compute the partial derivatives:

$$\frac{\partial\Phi}{\partial\rho} = \frac{\partial}{\partial\rho}(\rho^2 z) = 2\rho z, \qquad \frac{\partial\Phi}{\partial\phi} = 0, \qquad \frac{\partial\Phi}{\partial z} = \rho^2.$$

Hence,

$$\boxed{\nabla\Phi = 2\rho z\,\hat{\boldsymbol{\rho}} \;+\; 0\,\hat{\boldsymbol{\phi}} \;+\; \rho^2\,\hat{\mathbf{z}}}$$

(b) Line integral $\int_C \nabla\Phi \cdot d\mathbf{l}$ along the helix

The curve C is

$$\rho = a, \qquad \phi = t, \qquad z = bt, \qquad 0 \le t \le 2\pi.$$

In cylindrical coordinates, the differential line element is

$$d\mathbf{l} = d\rho\,\hat{\boldsymbol{\rho}} + \rho\,d\phi\,\hat{\boldsymbol{\phi}} + dz\,\hat{\mathbf{z}}.$$

Along C:

$$d\rho = 0, \quad \rho = a, \quad d\phi = dt, \quad dz = b\,dt \quad \Rightarrow \quad d\mathbf{l} = a\,dt\,\hat{\boldsymbol{\phi}} + b\,dt\,\hat{\mathbf{z}}.$$

Also along C, substitute $\rho = a$ and $z = bt$ into $\nabla\Phi$:

$$\nabla\Phi\big|_C = 2(a)(bt)\,\hat{\boldsymbol{\rho}} + a^2\,\hat{\mathbf{z}}.$$

Now take the dot product:

$$\nabla\Phi \cdot d\mathbf{l} = \left(2abt\,\hat{\boldsymbol{\rho}} + a^2\,\hat{\mathbf{z}}\right)\cdot\left(a\,dt\,\hat{\boldsymbol{\phi}} + b\,dt\,\hat{\mathbf{z}}\right) = a^2 b\,dt,$$

since $\hat{\boldsymbol{\rho}}\cdot\hat{\boldsymbol{\phi}} = 0$ and $\hat{\boldsymbol{\rho}}\cdot\hat{\mathbf{z}} = 0$.
Therefore,

$$\int_C \nabla\Phi \cdot d\mathbf{l} = \int_0^{2\pi} a^2 b\,dt = \boxed{2\pi a^2 b.}$$

(c) Verification (Fundamental Theorem of Line Integrals) and physical significance

Since the integrand is a gradient field, the fundamental theorem gives:

$$\int_C \nabla\Phi \cdot d\mathbf{l} = \Phi(\text{end}) - \Phi(\text{start}).$$

Start point ($t = 0$): $(\rho, \phi, z) = (a, 0, 0)$, hence

$$\Phi_{\text{start}} = a^2(0) = 0.$$

End point ($t = 2\pi$): $(\rho, \phi, z) = (a, 2\pi, 2\pi b)$, hence

$$\Phi_{\text{end}} = a^2(2\pi b) = 2\pi a^2 b.$$

Thus,

$$\Phi_{\text{end}} - \Phi_{\text{start}} = 2\pi a^2 b,$$

which matches part (b).

Physical significance: Because $\nabla\Phi$ is conservative, the line integral is *path-independent* and depends only on the endpoints. Interpreting Φ as a potential, $\int_C \nabla\Phi \cdot d\mathbf{l}$ equals the potential difference between the endpoints (and in mechanics/electrostatics it corresponds to work done by the associated conservative field).